

Test And Measurement EQUIPMENT REQUIREMENTS FOR DEFENCE

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This article discusses software-centric approach and other requirements in T&M equipment for defence services

The defence and aerospace (A&D) industry has developed hand in hand with the test equipment industry. The technology and electronics used in the test and measurement (T&M) industry are required to check for damage, quality, efficiency and reliability of the parts. Also, T&M equipment have assisted the growth of military and aerospace technology.

Avichal Kulshrestha, technical marketing engineer at National Instruments (NI), says that, the A&D industry finds itself in increasingly challenging times. It is under pressure to ensure timely execution and success of projects. Modern A&D electronics systems are entrusted with key goals that include protecting and equipping users with the best way possible and counter-evolving threats. These systems are also expected to work within the constraints of a challenging economy.

Kulshrestha adds that, T&M vendors globally are turning to a software-centric ecosystem with reconfigurable and modular hardware platforms with the objective to provide the A&D industry with a design-prototype-deploy-development chain. This approach to build test systems will ensure system stability, expandability and deployment over a long period of time.

Another key trend being observed is the need for higher bandwidth in modern test instruments for testing modern radars and electronic warfare (EW) systems. With growing bandwidth comes the requirement for high processing power, both of which are needed for performing critical inline computations during testing.

A&D technology is streamlined for speed, and it meets strict quality and accuracy demands for projects that stand the test of time. Evolving radio frequency (RF) requirements, obsolescence management and design for testing are some of the challenges that every test department in the defence industry aspires to curb.

NI's integrated toolchain and software-centric platform provides an innovative approach to quickly prototype a new design, improves component and systems testing and ensures system uptime and readiness. Awareness about the latest technologies can help deliver solutions with greater performance and reduced costs.

NI's customisable platform allows users to use the latest commercial technology and combine it with their own domain knowledge to build smarter test systems that meet present requirements and scale for the future.

The company recently released the second generation of its very popular vector signal transceiver device, VST 2.0. It is the first hardware in the industry with 1GHz instantaneous bandwidth. VST 2.0 also includes the latest Xilinx Virtex 7 series field programmable gate array (FPGA) for performing inline processing while testing

Fig. 1: National Instruments' vector signal transceiver device, VST 2.0
(Credit: www.ni.com)

